GATE (ALL BRANCHES)

Engineering Mathematics

Probability and Statistics





Lecture No. 02

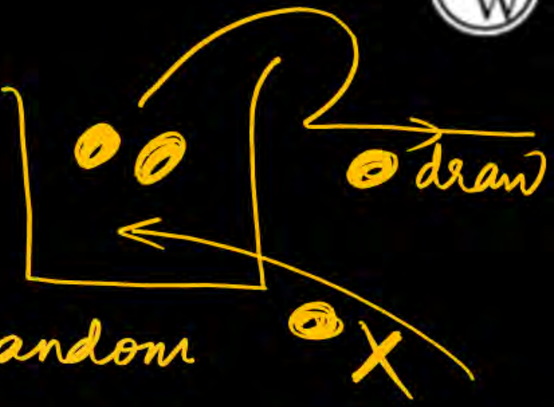
By- Rahul Sir



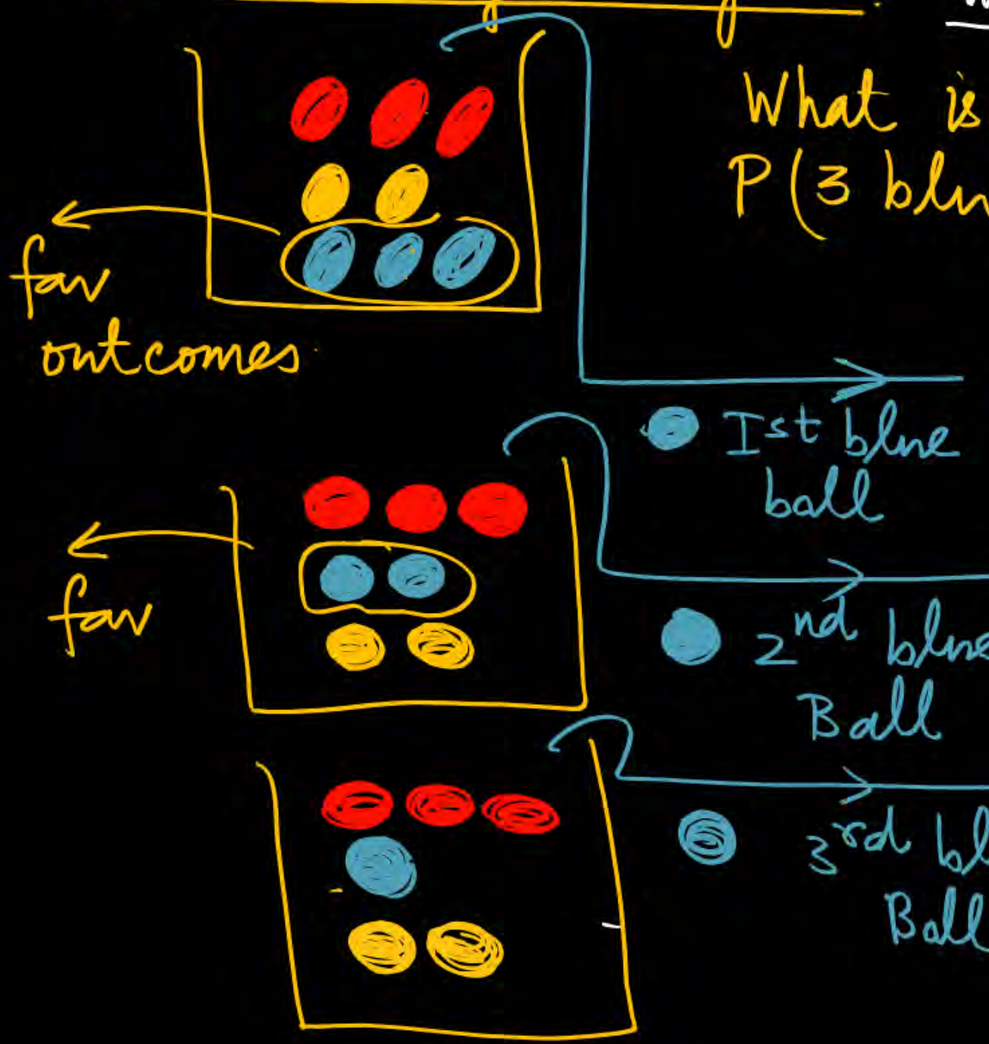
01 Fundamentals of Probability

02 Problem Solving Strategies

Counting strategies : without replacement :



What is the Probability
P(3 blue balls are drawn at random one at a time)



$$P(\text{blue } 1) = \frac{3}{8}$$

$$P\left(\frac{\text{blue } 2}{\text{blue } 1}\right) = \frac{2}{7}$$

$$P\left(\frac{\text{blue } 3}{\text{blue } 1, \text{blue } 2}\right) = \frac{1}{6}$$

W work



$= (W_1 \times W_2 \times W_3)$ ways

(→ together working)

$$P(\text{3 blue balls Are drawn at random}) = P(\text{blue 1})\, P\left(\frac{\text{blue 2}}{\text{blue 1}}\right) P\left(\frac{\text{blue 3}}{\text{blue 1, blue 2}}\right)$$

$$= \text{Working Together} = \text{Product Rule}$$

Product Rule

$$P(\text{3 blue balls}) = \boxed{\frac{3}{8} \times \frac{2}{7} \times \frac{1}{6}}$$

A) without replacement → Sample space Are Reduced

B) sample space — Prob change also

C) Dependent events — without replacement

$$P[\text{3 blue balls}] = \frac{6}{6 \times 7 \times 8} = \frac{1}{56}$$

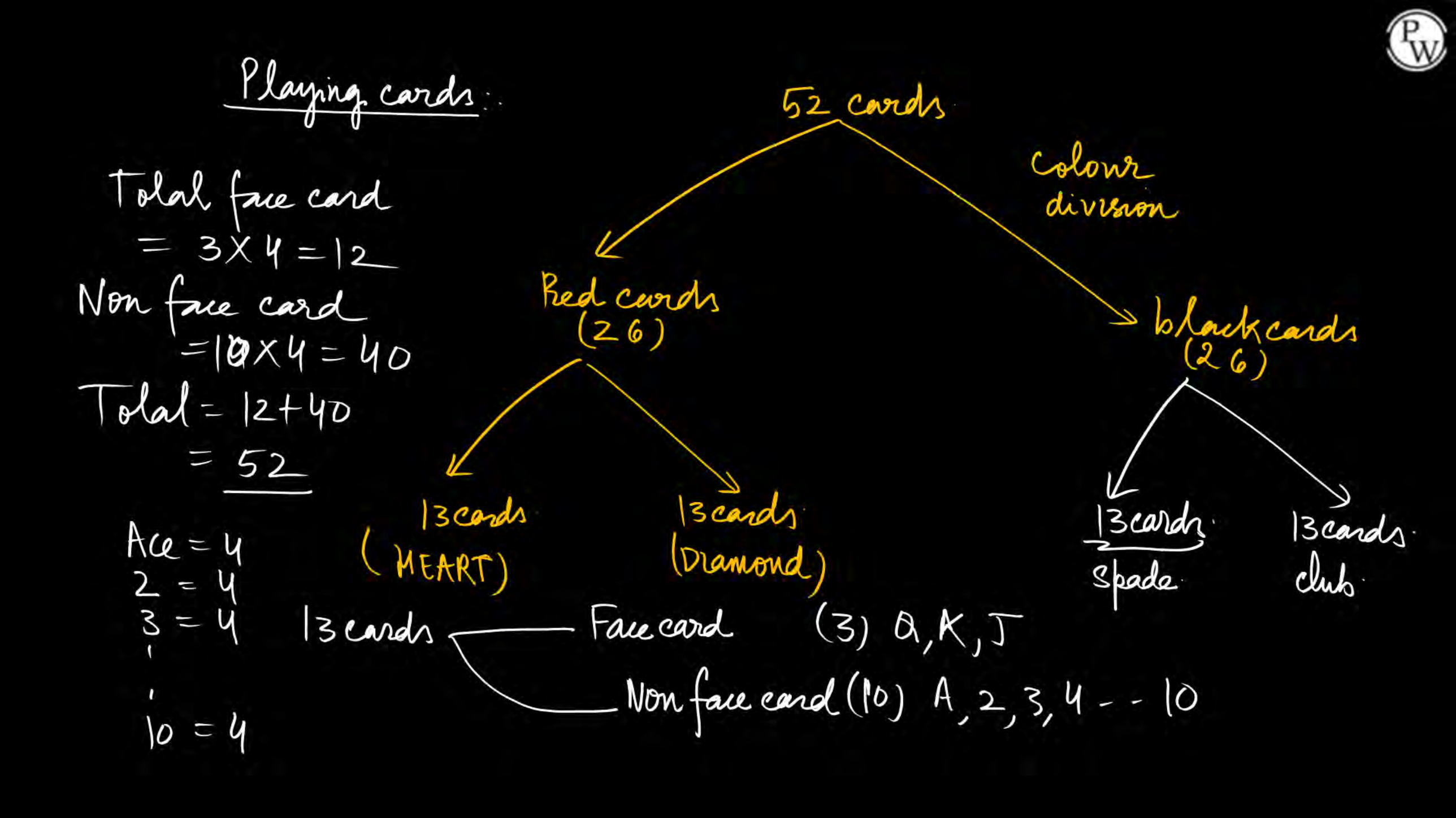
Playing cards:

Total face card
$= 3 \times 4 = 12$

Non face card
$= 10 \times 4 = 40$

Total $= 12 + 40$
$= 52$

Ace = 4
2 = 4
3 = 4
⋮
10 = 4



(without replacement)

If Three cards Are Drawn

First - King
Second - queen
Third - Jack

order fixed K, Q, J

4 King
4 Jack
4 Queen
52 cards

draw (1) → 4/52
1st King

3 King
4 Jack
4 queen
51 cards

SECOND draw → 4/51
IInd queen

3 King + 3 queen
4 Jack
50 cards

Third draw → 4/50

$$P(K \& J) = \frac{4}{52} \times \frac{4}{51} \times \frac{4}{50}$$ Ans

CASE(2)

Imp. case

4 King
4 queen
4 JACK
52 cards

I^st attempt ✓
J, Q, K

IInd attempt ✓
J, K, Q

IIIrd attempt ✓
K Q J

✓ IVth attempt → K J Q
✓ V^th attempt → Q J K
✓ VIth attempt → Q K J

T.ndividual work = +
Working Together = x]

If Three cards Are Drawn what is The Prob

" one of Those cards are Jack, queen, King

order - Not fixed

	✓	✓	✓	
A →	J	Q	K	+
B →	J	K	Q	+
C →	K	Q	J	+
D →	K	J	Q	+
E →	Q	J	K	+
F →	Q	K	J	+

$$P(E) = P(JQK) + P(JKQ) + P(KJQ) + P(KQJ) + P(QJK) + P(QKJ)$$

$$= \frac{4}{52} \times \frac{4}{51} \times \frac{4}{50} + \frac{4}{52} \times \frac{4}{51} \times \frac{4}{50} + \frac{4}{52} \times \frac{4}{51} \times \frac{4}{50} + \frac{4}{52} \times \frac{4}{51} \times \frac{4}{50} + \frac{4}{52} \times \frac{4}{51} \times \frac{4}{50} + \frac{4}{52} \times \frac{4}{51} \times \frac{4}{50}$$

$$= 6 \times \frac{4}{52} \times \frac{4}{51} \times \frac{4}{50} = \frac{4}{52} \times \frac{4}{51} \times \frac{4}{50} \times 3!$$

J Q K

3 different Items

n Different Items

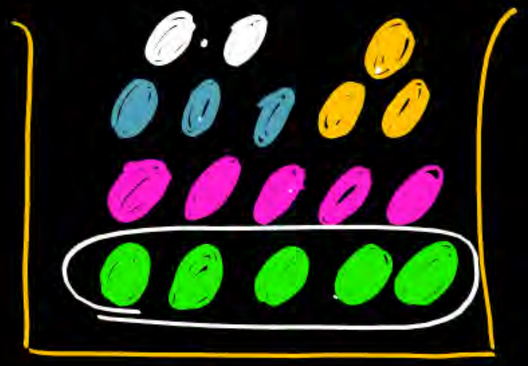
3	2	1

$= 3 \times 2 \times 1 = 6$ ways $= 3!$

3 boxes

n	n-1	n-2	n-3	n-4	1

$n \times (n-1) \times (n-2) \times (n-3) \times \cdots \times 1 = n!$ ways



What is the Prob
If 5 Balls are drawn at random one at a time
" one of those balls are green, Pink, blue, White yellow

A → G, P, B, W, Y

→ $\frac{5}{18} \times \frac{5}{17} \times \frac{3}{16} \times \frac{2}{15} \times \frac{3}{14}$

working Together

B → G, B, P, Y, W



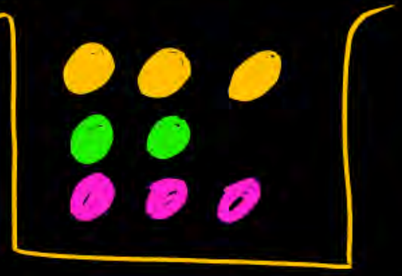
5	4	3	2	1
w_1	w_2	w_3	w_4	w_5

$w_1 \times w_2 \times w_3 \times w_4 \times w_5$

$P(GPBWY) = 5! \times \frac{5}{18} \times \frac{5}{17} \times \frac{3}{16} \times \frac{2}{15} \times \frac{3}{14}$

CASE (B) with replacement :



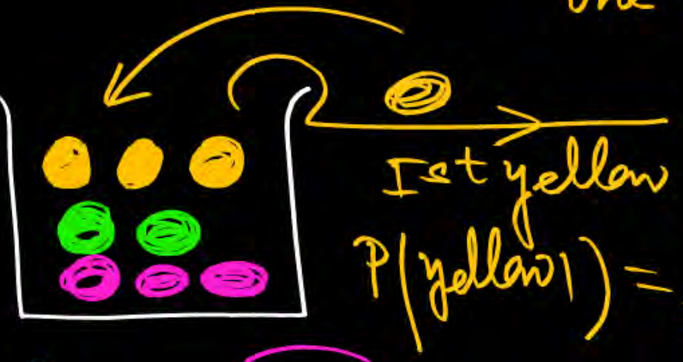


What is the Prob/chance

A) P(3 yellow balls Are drawn at random one at a time)

B) What is the Prob. If 3 balls Are drawn one of Those yellow, green, Pink.

(A)



$P(\text{yellow}_1) = \frac{3}{8}$



$P\left(\frac{\text{II yellow}}{\text{I yellow}}\right) = \frac{3}{8}$



$P\left(\frac{\text{yellow 3}}{1, 2}\right) = \frac{3}{8}$

$P(\text{3 yellow balls}) = \frac{3}{8} \times \frac{3}{8} \times \frac{3}{8}$

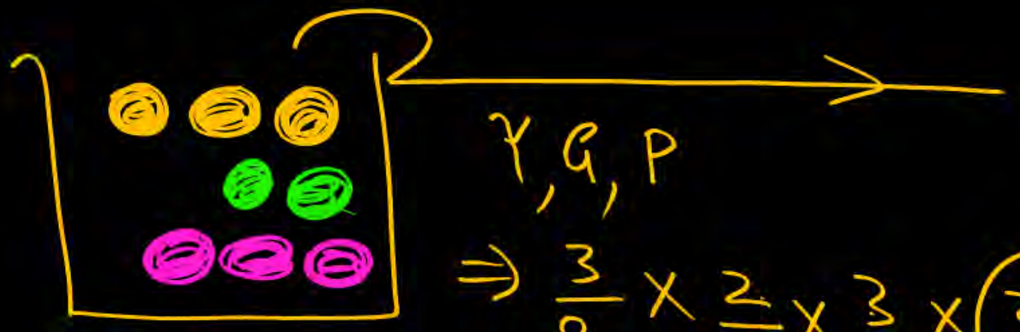
A) Chances don't change ⟶ prob. don't change

B) Sample space Not reduced

C) Independent events
(Prob remains const)

P(3 balls)
$= \frac{3}{8} \times \frac{3}{8} \times \frac{3}{8}$

Independent event ⟶ Next Prob. does Not depend in Previous one.

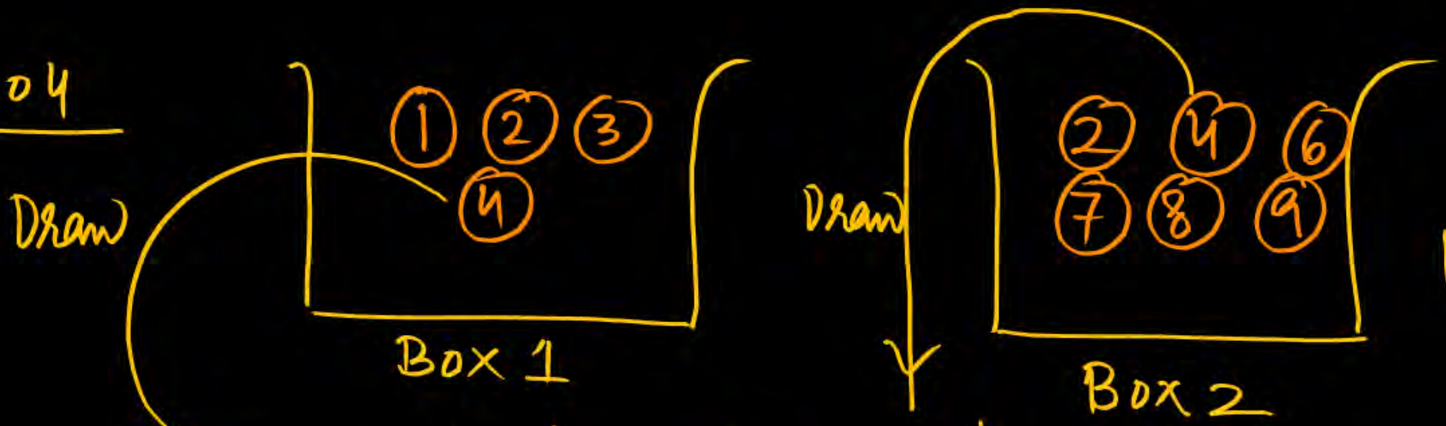


R, G, P

$\Rightarrow \frac{3}{8} \times \frac{2}{8} \times \frac{3}{8} \times 3!$

$= \frac{18 \times 3!}{8^3}$

✓ CASE 04



What is The P(TKt/coupon No 4)

$\frac{1}{4} + \frac{1}{6}$

Individual work

$$P(\text{Coupon 4}) = \frac{1}{4} + \frac{1}{6} = \frac{10}{24} = \frac{5}{12}$$

2) (What is The Prob. one box is Chosen and TKt 4 is drawn)

W_1 Job. W_2 Job



Working Ind

Box 1 | Box 2

$$\frac{1}{2} \times \frac{1}{4} + \frac{1}{2} \times \frac{1}{6}$$

$\frac{1}{2} \times \frac{1}{6}$ ↑ Working Together

Dependent

$$= \frac{1}{2} \times \frac{1}{4} + \frac{1}{2} \times \frac{1}{6}$$

$$= \frac{5}{24}$$

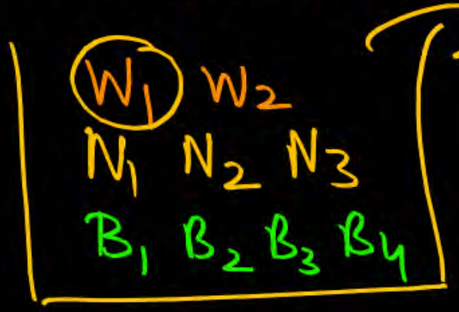
Q. Questions

$\frac{1}{1260}$

without replacement

A box contains 2 washers, 3 nuts and 4 bolts. Balls are drawn from the box randomly one at a time. Find the probability of drawing 2 washers first followed by 3 nuts and subsequently 4 bolts.

$W_1\ W_2\ N_1 N_2 N_3\ B_1 B_2 B_3 B_4$





$W_1\ W_2\ N_1 N_2 N_3\ B_1 B_2 B_3 B_4$

$\frac{2}{9} \times \frac{1}{8} \times \frac{3}{7} \times \frac{2}{6} \times \frac{1}{5} \times \frac{4}{4} \times \frac{3}{3} \times \frac{2}{2} \times \frac{1}{1}$

2W 3W 4 bolts

$= \frac{1}{1260}$ Ans

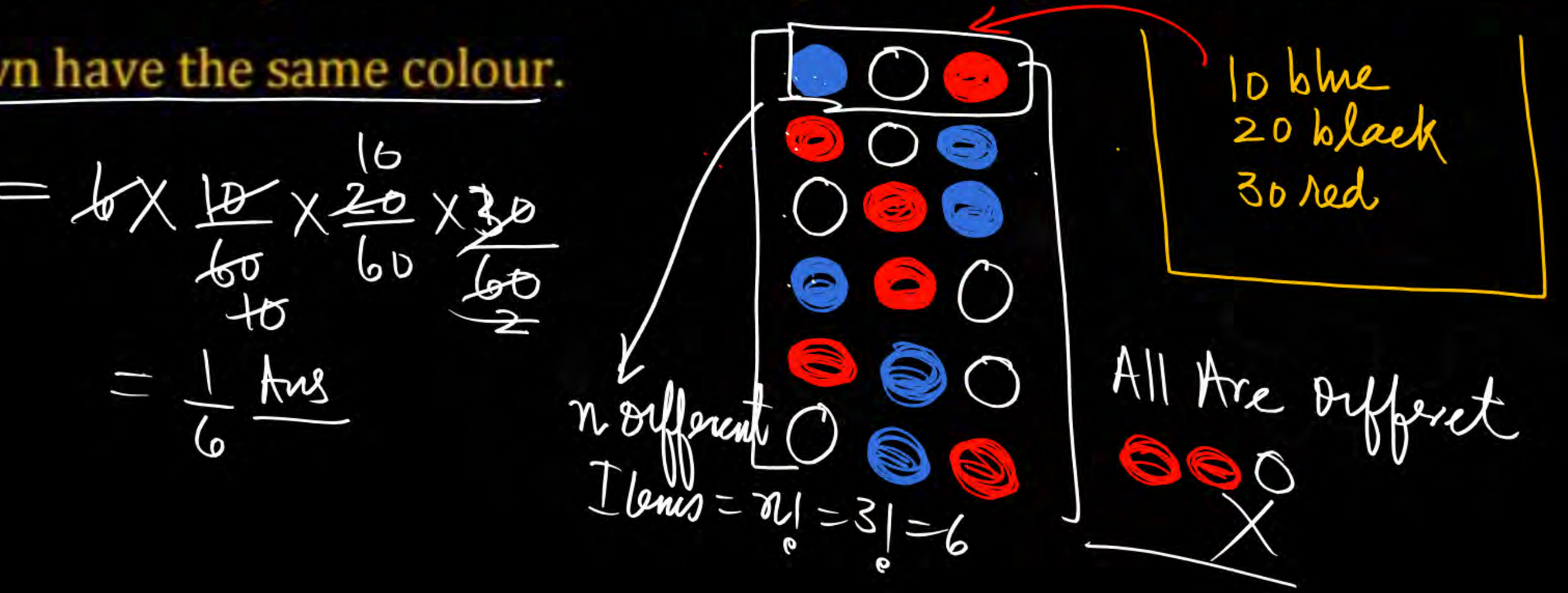
Q. Questions

with replacement

A bag contains 10 blue marbles, 20 black marbles and 30 red marbles. A marble is drawn from the bag, its colour is recorded and it is put back in the bag. This process is repeated 3 times. The probability that no two of the marbles drawn have the same colour.

$$= \cancel{6} \times \frac{\cancel{10}}{\cancel{60}_{\cancel{10}}} \times \frac{\overset{16}{\cancel{20}}}{60} \times \frac{\cancel{30}}{\cancel{60}_{2}}$$

$$= \frac{1}{6} \text{ Ans}$$



✓ Rahul Sir PW

Thank You!

GW Soldiers